4 Materials and Methods

Dataset generation

Random compositions of ternary NbMoTa refractory alloys with screw dislcoation are generated to compute the potential energy and Peierls barrier as follows. First pristine crystall strcuctures are generated by creating a rectangular simulation cell using the corresponding lattice constant and then randomly assigning solutes based on the desired concentrations. The simulation cell is oriented with dislocation glide direction x||[1,1,$\bar{2}$], glide plane normal direction, y||[$\bar{1}$10], and dislocation line direction, z||[111] with periodic boundary conditions along x and z and free surface along y. Atomic positions are relaxed by using a combination of the FIRE algorithm [97] and relaxation of the cell dimensions until the convergence is achieved-the norm of the force vector fell below 10e-6 eV/Å and stresses σ_{xx} , σ_{xz} , and σ_{zz} fell below 0.1 MPa. All the atomistic simulations are performed using LAMMPS [58] at zero temperature. A machine learning MTP potential is used to describe the interatomic interactions in the NbMoTa alloys [59, 34].

We then introduce a screw dislocation in the center of each relaxed simulation cell using the PAD method [16] and minimize the energy along with the relaxation of the pressures. This serves as our initial dislocated configuration. To generate the final dislocated configurations, we use the same initial pristine random structure and insert the dislocation at the adjacent position (next Peierls valley) relative to the initial dislocation at the distance of a along the glide direction where a is the Peierls valley distance. The potential energy changes is computed by subtracting the total potential energies of final and initial configurations. We then perform nudged elastic band (NEB) computations [60, 61, 62, 63] as implemented in LAMMPS [58] on these initial and final structures to compute the minimum energy path among these two energy states. The maximum value along this curve is stored as the corresponding Peierls barrier. The entire process is repeated for all the compositions in the training set (200 realizations for each composition) and test set (50 realizations for each composition) as depicted in Figure S1.

Graph representation

The results from atomistic simulations of atomic configurations and alloy properties are represented as graphs, which are used as inputs and labels to train the GNN. The graph structure is constructed from the pristine pure Mo configuration, where each Mo atom is a node in the graph, and the connectivity between nodes is determined by the distance between atoms. We consider a cylindrical region with a radius of $r_c = 16$Å, centered at the screw dislocation core, and only include atoms within this region as graph nodes. Within this graph, two atoms are connected if their distance is below the cutoff distance of 2.8 Å, which is the distance between nearest neighbors in a perfect crystal of molybdenum. The graph representation consists of 705 nodes and 2610 edges.

The node features contain information about chemicals (solute type) and structural defect (screw dislocation). The structural defect feature is constructed from the z-component of the displacement derived from the atomistic simulation in pure Mo. This feature remains identical for all the atomic configurations in the dataset, eleminating the need for atomic relaxation for new configurations during inference. The chemical feature of the node is uniqely determined for each composition represented as the solute types converted to one-hot encoded representations ([1,0,0], [0,1,0], and [0,0,1] for Nb, Mo, Ta, respectively). Moreover, the edge features representing the bond type are encoded as one-hot representations based on the neighboring nodes as shown in Table 1

Table 1: Edge feature representation for bond type connecting neighboring nodes, i and j.

node type i	**node type j**	**edge feature**
Nb	Nb	[1,0,0,0,0,0]
Mo	Mo	[0,1,0,0,0,0]
Ta	Ta	[0,0,1,0,0,0]
Nb (Mo)	Mo (Ta)	[0,0,0,1,0,0]
Nb (Ta)	Ta (Nb)	[0,0,0,0,1,0]
Mo (Ta)	Ta (Mo)	[0,0,0,0,0,1]